

Chapter one introduces the fundamental concepts of distributed systems and their trade-offs. The CAP theorem states that a distributed system cannot simultaneously guarantee consistency, availability, and partition tolerance. Engineers must choose which properties to prioritize.

Chapter two covers consensus algorithms including Paxos and Raft used in distributed databases. These algorithms ensure that all nodes in a cluster agree on the same value even in the presence of network partitions and node failures in production deployments.

Chapter three examines practical patterns for building resilient microservices architectures. Circuit breakers, bulkheads, and retry policies help services gracefully degrade under load. Observability through metrics, logs, and traces is essential for operating distributed systems.